

10. Aggregation and Indexing in MongoDB

Design and execute MongoDB queries using aggregation and indexing techniques with suitable examples.

```
gescoe@gescoe-OptiPlex-3010:~$ mongosh
Current Mongosh Log ID: 69ef1d9fb3c23f5fd7ce5f46
Connecting to:
  mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&ap
pName=mongosh+2.5.8
Using MongoDB: 8.2.1
Using Mongosh: 2.5.8
mongosh 2.8.2 is available for download: https://www.mongodb.com/try/download/shell
```

For mongosh info see: <https://www.mongodb.com/docs/mongodb-shell/>

```
-----
The server generated these startup warnings when booting
2026-04-27T13:15:13.656+05:30: Using the XFS filesystem is strongly recommended with the
WiredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
2026-04-27T13:15:17.230+05:30: Access control is not enabled for the database. Read and write
access to data and configuration is unrestricted
2026-04-27T13:15:17.230+05:30: For customers running the current memory allocator, we
suggest changing the contents of the following sysfsFile
2026-04-27T13:15:17.230+05:30: For customers running the current memory allocator, we
suggest changing the contents of the following sysfsFile
2026-04-27T13:15:17.230+05:30: We suggest setting the contents of sysfsFile to 0.
2026-04-27T13:15:17.230+05:30: We suggest setting swappiness to 0 or 1, as swapping can cause
performance problems.
-----
```

```
test> use Stud1;
switched to db Stud1
Stud1> db.createCollection("Stud");
{ ok: 1 }

Stud1>
db.Stud1.insertMany([ {Roll_No:1,Name:"Sita",Branch:"Comp",Marks:90},{Roll_No:2,Name:"Ra
m",Branch:"Comp",Marks:80},{Roll_No:3,Name:"Sham",Branch:"Comp",Marks:95},{Roll_No:4,
Name:"Radha",Branch:"IT",Marks:98},{Roll_No:5,Name:"Sam",Branch:"IT",Marks:70}]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ef1e2fb3c23f5fd7ce5f4c'),
    '1': ObjectId('69ef1e2fb3c23f5fd7ce5f4d'),
    '2': ObjectId('69ef1e2fb3c23f5fd7ce5f4e'),
    '3': ObjectId('69ef1e2fb3c23f5fd7ce5f4f'),
    '4': ObjectId('69ef1e2fb3c23f5fd7ce5f50')
  }
}
Stud1> db.Stud1.find();
[
  {
```

```

    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4c'),
    Roll_No: 1,
    Name: 'Sita',
    Branch: 'Comp',
    Marks: 90
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4d'),
    Roll_No: 2,
    Name: 'Ram',
    Branch: 'Comp',
    Marks: 80
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4e'),
    Roll_No: 3,
    Name: 'Sham',
    Branch: 'Comp',
    Marks: 95
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4f'),
    Roll_No: 4,
    Name: 'Radha',
    Branch: 'IT',
    Marks: 98
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f50'),
    Roll_No: 5,
    Name: 'Sam',
    Branch: 'IT',
    Marks: 70
  }
]
Stud1> db.Stud1.find().pretty();
[
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4c'),
    Roll_No: 1,
    Name: 'Sita',
    Branch: 'Comp',
    Marks: 90
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4d'),
    Roll_No: 2,
    Name: 'Ram',
    Branch: 'Comp',
    Marks: 80
  },
  {

```

```

    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4e'),
    Roll_No: 3,
    Name: 'Sham',
    Branch: 'Comp',
    Marks: 95
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4f'),
    Roll_No: 4,
    Name: 'Radha',
    Branch: 'IT',
    Marks: 98
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f50'),
    Roll_No: 5,
    Name: 'Sam',
    Branch: 'IT',
    Marks: 70
  }
]

```

```

Stud1> db.Stud1.aggregate([{$group : {_id : "$Branch", Total_Marks : {$sum : "$Marks"}}}]);
[ { _id: 'Comp', Total_Marks: 265 }, { _id: 'IT', Total_Marks: 168 } ]

```

```

Stud1> db.Stud1.aggregate([{$group : {_id : "$Branch", Minimum_Marks : {$min :
"$Marks"}}}]);
[
  { _id: 'IT', Minimum_Marks: 70 },
  { _id: 'Comp', Minimum_Marks: 80 }
]

```

```

Stud1> db.Stud1.aggregate([{$group : {_id : "$Branch", Avg_Marks : {$avg : "$Marks"}}}]);
[
  { _id: 'Comp', Avg_Marks: 88.33333333333333 },
  { _id: 'IT', Avg_Marks: 84 }
]

```

```

Stud1> db.Stud1.aggregate([{$group : {_id : "$Branch", Maximum_Marks : {$max :
"$Marks"}}}]);
[
  { _id: 'Comp', Maximum_Marks: 95 },
  { _id: 'IT', Maximum_Marks: 98 }
]

```

```

Stud1> db.Stud1.aggregate([{$group : {_id : "$Branch", Total_Count : {$count : {}}}}]);
[ { _id: 'IT', Total_Count: 2 }, { _id: 'Comp', Total_Count: 3 } ]

```

```

Stud1> db.Stud1.aggregate([{$match: {Branch: "Comp"}}]);
[
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4c'),
    Roll_No: 1,
    Name: 'Sita',
    Branch: 'Comp',

```

```

    Marks: 90
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4d'),
    Roll_No: 2,
    Name: 'Ram',
    Branch: 'Comp',
    Marks: 80
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4e'),
    Roll_No: 3,
    Name: 'Sham',
    Branch: 'Comp',
    Marks: 95
  }
]

```

```
Stud1> db.Stud1.aggregate([{$project: {Roll_No:1,Name:1}}]);
```

```

[
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4c'),
    Roll_No: 1,
    Name: 'Sita'
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4d'),
    Roll_No: 2,
    Name: 'Ram'
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4e'),
    Roll_No: 3,
    Name: 'Sham'
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4f'),
    Roll_No: 4,
    Name: 'Radha'
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f50'),
    Roll_No: 5,
    Name: 'Sam'
  }
]

```

```
Stud1> db.Stud1.aggregate([{$sort: { Marks: 1 }}]);
```

```

[
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f50'),
    Roll_No: 5,

```

```

    Name: 'Sam',
    Branch: 'IT',
    Marks: 70
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4d'),
    Roll_No: 2,
    Name: 'Ram',
    Branch: 'Comp',
    Marks: 80
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4c'),
    Roll_No: 1,
    Name: 'Sita',
    Branch: 'Comp',
    Marks: 90
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4e'),
    Roll_No: 3,
    Name: 'Sham',
    Branch: 'Comp',
    Marks: 95
  },
  {
    _id: ObjectId('69ef1e2fb3c23f5fd7ce5f4f'),
    Roll_No: 4,
    Name: 'Radha',
    Branch: 'IT',
    Marks: 98
  }
]

```

```

Stud1> db.Stud1.createIndex({"Roll_No":1});
Roll_No_1

```

```

Stud1> db.Stud1.getIndexes();
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { Roll_No: 1 }, name: 'Roll_No_1' }
]

```

```

Stud1> db.Stud1.dropIndexes();
{
  nIndexesWas: 2,
  msg: 'non-_id indexes dropped for collection',
  ok: 1
}
Stud1> db.Stud1.getIndexes();
[ { v: 2, key: { _id: 1 }, name: '_id_' } ]

```